PQC DAILY STUDY SERIES

The Impact of Post-Quantum Cryptography on AI and Machine Learning Security

Day 54 | Category: PQC Foundational Concepts Date: 2026-11-12 | DAILY MONITORING

PQC's Intersection with AI and Machine Learning

As both Post-Quantum Cryptography (PQC) and Artificial Intelligence (AI) mature, their intersection is becoming a critical focal point for cybersecurity. While PQC aims to secure data from future quantum computers, AI and Machine Learning (ML) models are themselves valuable assets that require robust protection. The very data that trains these models, and the models themselves, are vulnerable to quantum threats.

The Core Problem: Data and Model Vulnerability

AI/ML systems heavily rely on the confidentiality and integrity of vast datasets during training and inference. A quantum computer could break the classical cryptography (like RSA and ECC) used to protect this data, leading to several catastrophic scenarios:

- **Data Poisoning:** An attacker could decrypt sensitive training data, subtly manipulate it, and re-encrypt it. The compromised data would then be used to train an ML model, creating backdoors or biased outputs that serve the attacker's goals.
- **Model Theft:** Many proprietary ML models are high-value intellectual property. An attacker with a quantum computer could intercept and decrypt a model during transmission or from storage, stealing years of research and investment.
- **Inference Attacks:** Data sent to a remote ML model for inference (e.g., a medical diagnosis query) is also protected by classical encryption. A quantum attack could expose this sensitive user data.

Securing the AI/ML Lifecycle with PQC

Integrating PQC is essential for providing long-term security for AI/ML systems. This involves securing each stage of the ML lifecycle against quantum adversaries.

PQC Integration Points in the ML Lifecycle

ML Lifecycle Stage	Classical Security Method	Post-Quantum Solution	Impact
Data Collection & Storage	AES-256 for data at rest, TLS (using RSA/ECC) for data in transit.	AES-256 remains secure. TLS must be upgraded with a hybrid scheme (e.g., X25519+Kyber768).	Protects sensitive training data from "Harvest Now, Decrypt Later" attacks.
Federated Learning	Secure Aggregation using classical commitments and signatures.	Use of post-quantum digital signatures (like ML-DSA) to verify the integrity and origin of model updates from participants.	Prevents a quantum-capable attacker from forging malicious model updates and poisoning the global model.
Model Distribution & Updates	Code signing with RSA or ECDSA.	Firmware and model files must be signed with ML-DSA or SLH-DSA.	Ensures the authenticity and integrity of deployed models, preventing the distribution of compromised or backdoored AI systems.

Furthermore, emerging concepts like "Homomorphic Encryption" (which allows computation on encrypted data) are also being researched in post-quantum variants. A post-quantum homomorphic encryption scheme would allow an ML model to be trained on data that is never decrypted, providing an unprecedented level of privacy and security against all attackers, classical and quantum.

Daily Quiz

1. What is the primary risk a quantum computer poses to the training data of an ML model?
2. How can ML-DSA (CRYSTALS-Dilithium) be used to secure the Federated Learning process?
3. Why is simply using AES-256 for data at rest insufficient for providing comprehensive security for an entire AI/ML pipeline?